



NCHRP 08-154

June 2023 | Naomi Stein

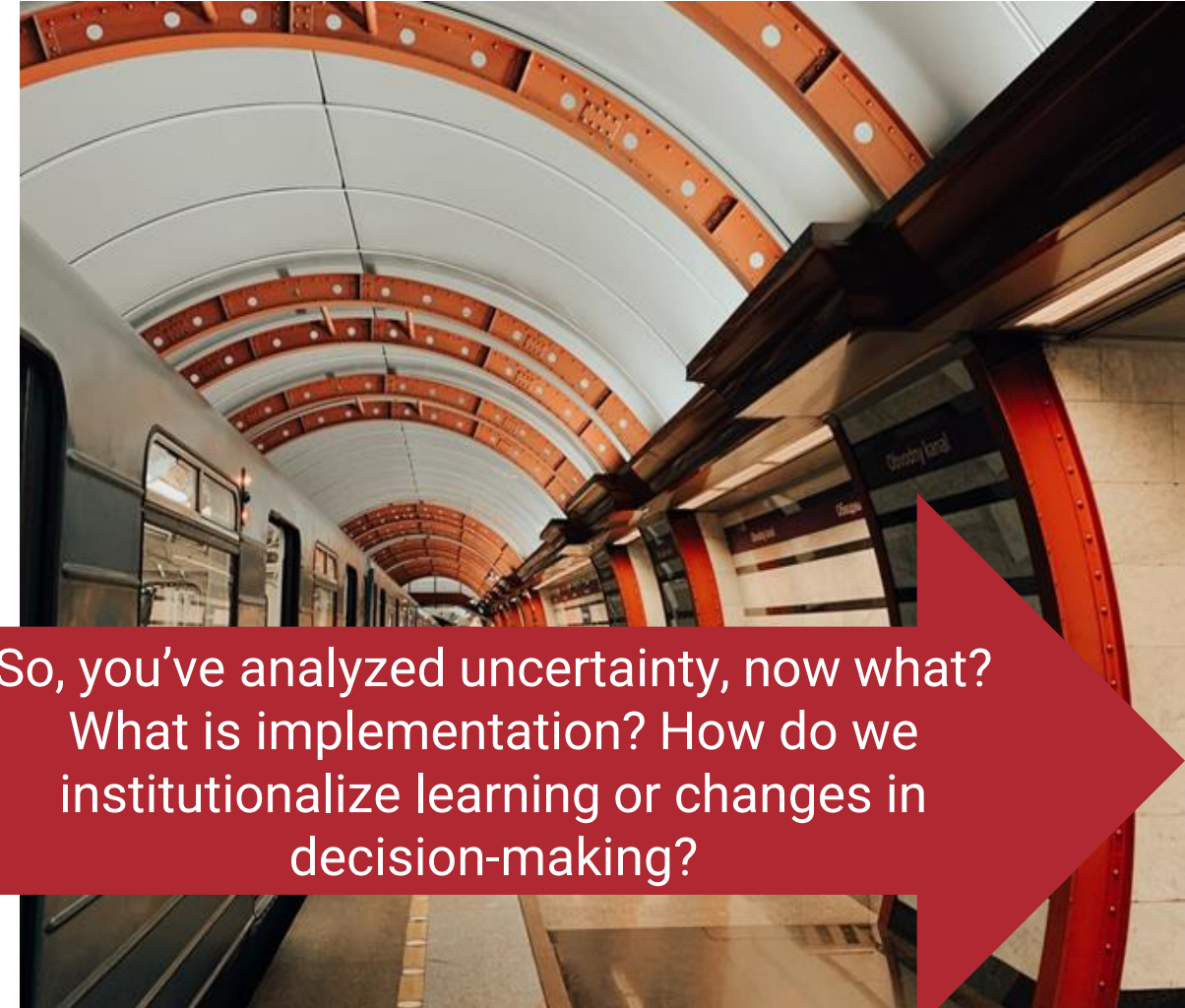
ITAP Workshop

Uncertainty in transportation planning and programming



CONTEXT: **WHERE WE'RE STARTING**

- State DOTs and MPOs are facing both new and compounding uncertainties
- Federally mandated planning process presents opportunities and constraints
- Increased focus on performance targets and outcomes
- Advances in exploring drivers of uncertainty with methods such as scenario planning, robust decision-making, risk analysis



So, you've analyzed uncertainty, now what?
What is implementation? How do we
institutionalize learning or changes in
decision-making?

WHAT WILL THE PROJECT ACCOMPLISH AND FOR WHOM?



ENABLE DECISION-MAKERS AND STAFF WITHIN STATE DOTs AND MPOs

to identify key points of flexibility within the planning process to better address uncertainty.



PROVIDE TECHNICAL AND ORGANIZATIONAL STRATEGIES

to inform investment decisions and proactive adaptation to meet agency goals.

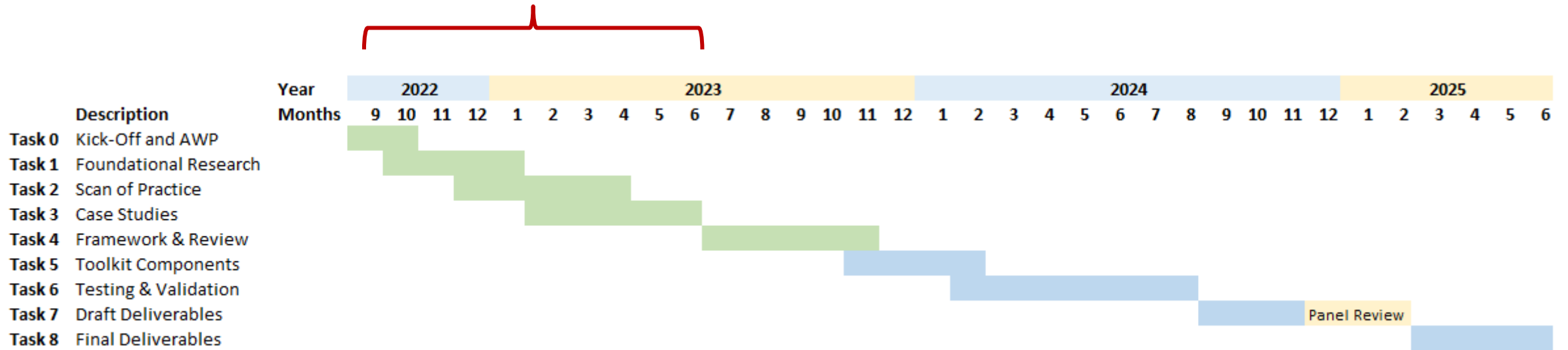


Core Audience: DOT and MPOs – Why, When, Where, and How to Consider & Manage Uncertainty

- Leadership → More General – Focus on Making the Case and Providing Roadmap for Proactive Adaptation
- Technical Staff → Specific Strategies, Resources – Integrating into Business Process

Secondary Audience: Support agency staff as they communicate with stakeholders, including partners, decision-makers, elected officials, and the public-at-large about uncertainty in transportation planning and programming.

WE ARE HERE: FACT FINDING



PROJECT SCHEDULE

We are wrapping up our first phase of work.

Fact Finding



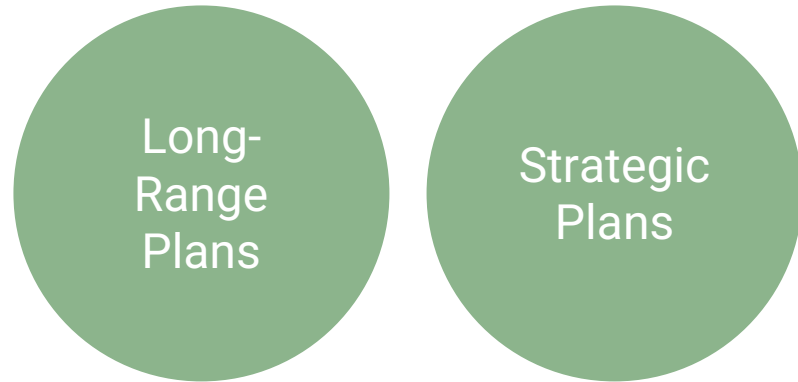
- Literature and practice review
- Scan of practice through questionnaire
 - Majority State DOTs
 - Other responses from MPOs/RPOs, Transit Agencies, Local Government, and others
 - Mix of roles (e.g., planning, forecasting, analysts, managers, executives)
- Focus groups with industry representatives and researchers from outside the typical transportation agency context
- *General themes identified to date on the following slides – more details to come!*

Important Sources of Uncertainty Identified in Outreach

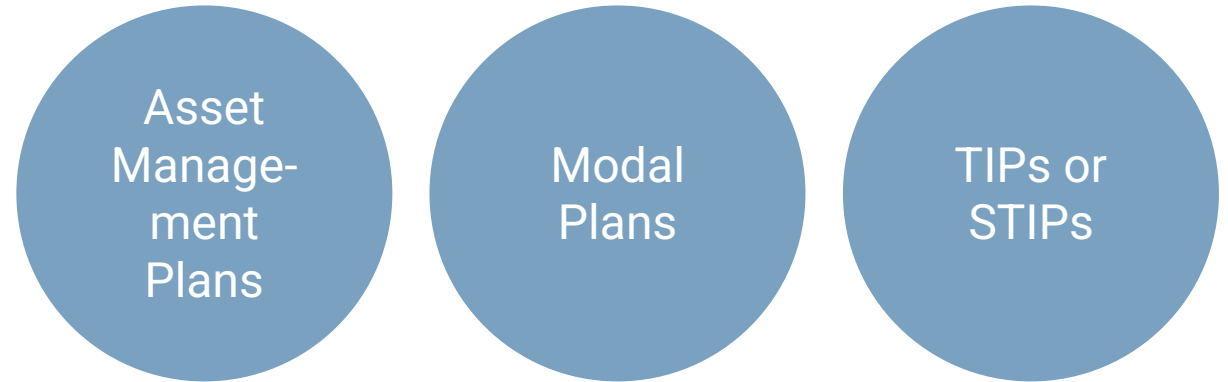


Other sources: mode choice, safety tech/behavior, vehicle electrification, telework, workforce, land use, vehicle automation, location choice, e-commerce, other disruptions

MORE OFTEN ADDRESSED WITHIN:



UNCERTAINTY ALSO ADDRESSED IN OTHER PLANS:

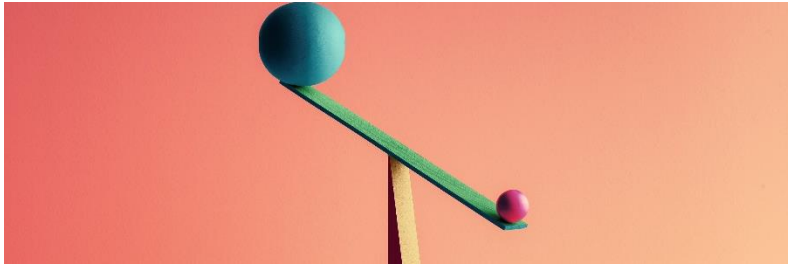


Commonly addressed:

Funding, policies and regulations, costs, climate/natural/environmental hazards

Also addressed:

Mode choice, vehicle electrification, safety tech/behavior, automation, economic change, land use, workforce, other disruptions, location choice, telework



Methods

Scenario Planning
Risk Management
Sensitivity Analysis
Contingency Factors
Tradeoff and Exploratory Analysis
Causal Diagramming
Monte Carlo

Tools

Travel Demand Model
Pavement Management System
Bridge Management System
Risk and Vulnerability Screening/Assessment Tools
Visioning/Sketch Planning
Transit Asset Management System

Barriers and Challenges

- Complexity of coordination and communication
- Inertial effects – challenge to adopt new things
- Lack of funding and resources (data, staff, tools)
- Compounding or deep uncertainty – “overwhelm”

Needs

- Tools
- Updated data sources
- Shared best practices
- Collaboration and communication across organizations
- Funding and staff resources
- Project management
- Visualization tools



Themes - To Get Your Wheels Turning

Build and leverage partnerships

Interact with other industries (tech, energy)

Share rather than "recreate the wheel"

Mediate risk of being "first adopters"

Focus on outreach, education, and communication

Resistance to change or uncertainty is common

Share successes, peer examples

Challenges: silos, differences in terminology

Adapt and prepare for change

Focus on flexibility

Modularity, project "on ramps" and "off ramps"

Tracking to identify tipping points or need to course correct

Use up to date data and projections

Remember the average can obscure the "cutting edge"

Incorporate long-range lessons in stress tests

Move the locus of certainty from a prediction
(I am certain that the prediction is accurate)
to the plan
(I am certain that my plan is robust in the face of many futures).